

Model Comparison Report

Logistic Regression vs. Support Vector Machine (SVM)

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1. Overview

This assignment extends the hotel booking cancellation analysis by training and comparing two classification models: Logistic Regression (baseline from Assignment 2) and a Support Vector Machine (SVM) with a balanced class weight. Both models are evaluated on the same 74-record held-out test set, using the same preprocessing pipeline (StandardScaler for numeric features; OneHotEncoder for categorical features).

The target variable is Booking_Cancelled (0 = No, 1 = Yes). The test set contains 69 non-cancellations and 5 actual cancellations, a highly imbalanced ratio of approximately 14:1.

2. Confusion Matrices

2.1 Logistic Regression: Confusion Matrix



True Negatives (67): Guests correctly predicted as non-cancellers. The model excels here, capturing 97% of the majority class.

False Positives (2): Two guests were incorrectly flagged as likely cancellers. In practice, this triggers unnecessary retention effort for guests who intended to stay.

False Negatives (3): Three real cancellations the model completely missed. These are the most costly errors, lost room revenue with no opportunity to intervene.

True Positives (2): Two cancellations were correctly caught, representing a 40% detection rate on the class that matters most.

2.2 SVM: Confusion Matrix

Classification Report:				
	precision	recall	f1-score	support
No Cancelled	0.96	0.97	0.96	69
Cancelled	0.50	0.40	0.44	5
accuracy			0.93	74
macro avg	0.73	0.69	0.70	74
weighted avg	0.93	0.93	0.93	74

True Negatives (66): One fewer correct non-cancellation prediction compared to Logistic Regression, SVM incorrectly flagged one extra non-cancelling guest.

False Positives (3): Three guests were wrongly predicted as cancellers, up from 2 in Logistic Regression. The class_weight='balanced' setting caused the SVM to be more aggressive in predicting the minority class.

False Negatives (3): Identical to Logistic Regression, SVM also missed 3 of 5 real cancellations, despite its balanced class weighting.

True Positives (2): Same as Logistic Regression, the SVM caught the same 2 cancellations, yielding a 40% recall on the cancellation class.

3. Evaluation Metrics: Detailed Results

The table below presents per-class metrics from both classification reports side by side.

Metric	LR: No Cancel	LR: Cancelled	SVM: No Cancel	SVM: Cancelled
Precision	0.96 (96%)	0.50 (50%)	0.96 (96%)	0.40 (40%)
Recall	0.97 (97%)	0.40 (40%)	0.96 (96%)	0.40 (40%)
F1-Score	0.96 (96%)	0.44 (44%)	0.96 (96%)	0.40 (40%)
Accuracy	93.24%	,	91.89%	,

3.1 Accuracy

Logistic Regression achieved 93.24% overall accuracy versus SVM's 91.89%. Both figures are inflated by the dominant majority class; a model predicting 'No Cancel' for every record would score 93.2% without ever detecting a single cancellation. Accuracy is, therefore, a poor primary metric in this imbalanced setting.

3.2 Precision (Cancellation Class)

Logistic Regression achieved a precision of 0.50 on the cancellation class; SVM achieved 0.40. Logistic Regression is more reliable when it does flag a cancellation, 50% of its cancellation predictions are correct compared to only 40% for SVM.

3.3 Recall (Cancellation Class)

Both models achieved a recall of 0.40 on the cancellation class, catching only 2 of 5 real cancellations. Despite SVM using `class_weight='balanced'`, specifically designed to improve minority-class detection, it did not recover the 3 missed cancellations that Logistic Regression also failed to catch. This suggests the dataset is too small (only 5 cancellations in the test set) for balancing alone to make a meaningful difference.

3.4 F1-Score (Cancellation Class)

The F1-score, the harmonic mean of precision and recall, summarises overall minority-class performance. Logistic Regression scores 0.44 versus SVM's 0.40. The difference arises entirely from LR's higher precision (0.50 vs 0.40), since recall is identical for both.

4. Head-to-Head Comparison

Metric	Logistic Regression	SVM
Accuracy	93.24%	91.89%
Precision (Cancelled)	0.50	0.40
Recall (Cancelled)	0.40	0.40
F1-Score (Cancelled)	0.44	0.40
Most Important Metric (Recall)	0.40	0.40

Logistic Regression outperforms SVM on three of the four key metrics (accuracy, precision, and F1-score for the cancellation class). Both models tie on recall, the single most important metric for this business problem, catching only 40% of real cancellations.

The SVM's balanced class weighting increased false positives (3 vs 2) without improving true positive detection, resulting in lower precision and a lower F1-score. This trade-off was not worthwhile given the dataset size.

5. Which Metric Matters Most and Why

For hotel revenue management, recall on the cancellation class is the most important metric. The business consequence matrix is asymmetric:

Error Type	What Happened	Business Cost
False Negative	The model missed a cancellation; the guest canceled without warning	HIGH, empty room, lost revenue, no time to intervene or re-sell
False Positive	The model flagged a non-cancelling guest as a risk	LOW, unnecessary outreach or discount offer to a guest who stays

A recall of 0.40 means the model fails to catch 60% of real cancellations. In a hotel with higher cancellation volumes, this represents substantial unmitigated revenue risk. Maximising recall, even at the cost of triggering more false alarms, should be the primary optimisation objective in future model iterations.

6. Limitations and Recommendations

6.1 Limitations

- Small test set: Only 5 actual cancellations appear in the 74-record test set, making metric estimates highly variable. A single correct or incorrect prediction shifts recall by 20 percentage points.
- Temporal/seasonal bias: If data were collected during a specific season (e.g., peak summer), cancellation patterns may not generalise to other periods.
- Static model: Neither model accounts for real-time context, promotions, external events, or economic conditions that a human operations manager would consider.
- Balanced weighting did not help: SVM's `class_weight='balanced'` increased false positives without recovering missed cancellations, suggesting the signal in the features is insufficient rather than the learning algorithm.

6.2 Recommendations

- Collect more data: A larger dataset with more cancellation examples is the single most impactful improvement available.
- Apply SMOTE: Synthetic oversampling of the minority class during training would provide a more balanced gradient signal.
- Threshold tuning: Lowering the decision threshold below 0.5 on the Logistic Regression model would increase recall at a controlled cost to precision.
- Explore tree-based models: Random Forest and Gradient Boosting often outperform linear classifiers on tabular data with categorical features.
- Cross-validation: With a dataset this small, k-fold cross-validation would give more reliable performance estimates than a single train/test split.